

TQDC16VS-E Data Format

Hardware TQDC16VS-E

TQDC16VS-E payload data is encapsulated in M-Stream 2.2 protocol using Data Subtype 0.

M-Stream Header			
word #	byte offset	bits	description
0	0	31:24	Device ID
		23:18	Flags
		17:16	Subtype
		15:0	Fragment length, 4*(N-2)
1	4	31:16	Packet ID
		15:0	Fragment offset (bytes)
M-Stream Subtype 0, first fragment			
2	8	31:0	Device Serial
3	12	31:24	reserved
		23:0	Event number
4	16	31:0	Event timestamp, TAI seconds
5	20	31:2	Event timestamp, TAI nanoseconds
		1:0	Event timestamp, TAI flags
6	24	31:0	Event Data
...			
N-1	4*(N-1)	31:0	
M-Stream Subtype 0, next fragments			
2	8	31:0	Event Data
...			
N-1	4*(N-1)	31:0	

Event Data format

word #	byte offset	bits	description
0	0	31:0	
...			
S ₁ /4-1	S ₁ -4	31:0	

$S_1/4$	S_1	31:0	Data Block #2 (S_2 bytes)
...			
$(S_1+S_2)/4-1$	S_1+S_2-4	31:0	
...			
Till the last data block			
...			

Data Block format

word #	byte offset	bits	description
0	0	31:28	Data Type
		27:24	Channel number
		23:19	reserved
		18:16	ADC Data Block specific
		15:0	Data Payload length (4*(N-1) bytes)
1	4	31:0	Data Payload
...			
N-1	4*(N-1)	31:0	

TDC Data Block
ADC Data Block

TDC Data Block

TDC data type is 0x0.
channel number field not used

TDC Data Payload

2 - TDC event header																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
2				TDC ID				Event number													TDC timestamp										

TDC timestamp in number of 25 ns time intervals since global trigger timestamp found in event header.

3 - TDC event trailer																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

3	TDC ID	Event number	TDC word count
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4 - TDC data, leading edge																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
4				reserved				channel				data																			

5 - TDC data, trailing edge																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
5				reserved				channel				data																			

TDC time measurement

data bits 20:2 = number of 100 ps time intervals since global trigger timestamp found in event header;
data bits 1:0 = rcdata[1:0].

6 - TDC error																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
6				TDC ID				reserved								TDC error flags															

TDC error flags bits:

- [0] Hit lost in group 0 from read-out fifo overflow
- [1] Hit lost in group 0 from L1 buffer overflow
- [2] Hit error have been detected in group 0
- [3] Hit lost in group 1 from read-out fifo overflow
- [4] Hit lost in group 1 from L1 buffer overflow
- [5] Hit error have been detected in group 1
- [6] Hit lost in group 2 from read-out fifo overflow
- [7] Hit lost in group 2 from L1 buffer overflow
- [8] Hit error have been detected in group 2
- [9] Hit lost in group 3 from read-out fifo overflow
- [10] Hit lost in group 3 from L1 buffer overflow
- [11] Hit error have been detected in group 3
- [12] Hits rejected because of programmed event size limit
- [13] Event lost (trigger FIFO ovefrlow)
- [14] Internal fatal chip error has been detected

Most important error bits are 12 and 13. Error bit 14 should be ignored.

For detailed description of TDC please refer to HPTDC manual [↗](#).

ADC Data Block

ADC data type is 0x1.

ADC Data Block specific bits:

- [16] - reserved
- [17] - ADC header FIFO overflow (mfifo)
- [18] - reserved

Each ADC data block may include several signals. Each signal data starts with header followed by digitizer data.

ADC header																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADC data length (in bytes)																ADC Timestamp															

ADC timestamp is number of 8 ns intervals after event timestamp.

ADC Data																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Word N+1																Word N															

~~If number of ADC words is odd, then bits [31:16] of last ADC data payload word should be 0x0000.~~

ADC data aligned to MSB.

RawData
